BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A (MCQ's)

Q1. Numerical value of _____ remains constant every where:

(a) g
(b) G
(c) F
(d) W

Q2. Spring balance is used to measure:

(a) Mass
(b) Weight
(c) Elasticity
(d) Density

Q3. Your weight as measured on Earth will be _____ on Moon.

(a) Increased
(b) Decreased
(c) Remains same
(d) None of these

Q4. The velocity of a satellite is _____ of its mass.

(a) Independent
(b) Dependent
(c) Equal
(d) Double

Q5. During which process a gas becomes a liquid:

(a) Melting
(b) Freezing
(c) Condensing
(d) Boiling

Q6. The energy released during fission of fusion reaction is called:

(a) Solar energy
(b) Geothermal energy
(c) Tidal energy
(d) Nuclear energy

Q7. Which of the following material is more elastic?

(a) Rubber
(b) Glass
(c) Steel
(d) Wood

Q8. Which is the unit for the spring constant?

(a) N.m
(b) N.m2
(c) N.m
(d) N.m2

Q9. Which of the following is not a unit of pressure?

(a) Pascal
(b) Bar
(c) Atmosphere
(d) Newton

Q10. Newton's law of gravitation holds between every two objects on the:

(a) Earth
(b) Jupiter
(c) Moon
(d) Universe

Q11. Gravitational field of Earth is directed:

(a) Towards the Earth
(b) Towards the Sun
(c) Towards the Moon
(d) Away from Earth

Q12. _____ was the first scientist who gave the concept of gravitation:

(a) Einstein
(b) Newton
(c) Faraday
(d) Maxwell

Q13. Quantity of motion contained in a body is called:

(a) Force
(b) Inertia
(c) Momentum
(d) Gravity

Q14. Cntrifugal force is always directed:

(a) Towards centre
(b) Away from centre
(c) Along the circulr path
(d) All sides

Q15. A pair of inlike parallel forces having different lines forces produce:

(a) Equilibrium
(b) Torque
(c) A couple
(d) Unstable equilibrium

Q16. Head to tail rule can be used to add _____ forces.

(a) Two
(b) Three
(c) Five
(d) Any number of

Q17. A body is in equilibrium when it has:

(a) Uniform speed
(b) Uniform acceleration
(c) Both a and b
(d) Zero acceleration

Q18. A body is in neutral equilibrium when its centre of gravity:

(a) Is at the lowest position
(b) Remains at same height
(c) Is at highest position
(d) Is at its base

Q19. A world wide system of measurements in which the units of base quantities were introduced is called _____

(a) Prefixes
(b) International system of units
(c) Hexadecimal system
(d) None of these

Q20. Length, mass, electric current, time intensity of light and amount of substance are examples of:

(a) Base quantities
(b) Derived quantities
(c) Prefixes
(d) Quartile quantities

Section-B

> Note: Answer any EIGHT of the following question. Each question carries 03 marks.

Q2. Derive the equation S = vit + 1/2at2

Q3. Differentiate between distrace and displacement.

Q4. A bus is moving on a road with 15 m sand it accelerates at 5 ms2. Find the final velocity of bus after 6 seconds.

Q5. Describe friction and write the types of friction.

Q6. A force of 3400 N is applied on a boyd of mass is 850 k.g Find the acceleration produced by the force.

Q7. Define couple as a pair of forces tending to produce torque.

Q8. State Hooke's Law.

Q9. What is energy? Name the different forms of energy.

Q10. Convert 30oC into Kelvin and Fahrenheit scale.

Q11. Write a note on renewable energy sources.

Q12. Define like and unlike forces.

Section-C (Descriptive Answer)

> Note: Ansewr any TWO of the following question. Each question carries 06 marks.

Q13. Define momentum. Explain the law of conservation fo momentum with the help of example.

Q14. Wate and explain Newton's Law of gravitation.

Q15. Write notes on any TWO of the following: (a) State of equilibrium (b) Newton's Third Law of motion. (c) Mass fo Earth